

Machine Learning – Chapter 3

Basic Concepts of Learning | S.
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Chapter 3: Basic Concepts of Learning

Learning: improving performance through experience

Concept learning and hypothesis spaces

Generalization and specialization

Inductive bias

Design of a learning system

Learning frameworks: PAC, mistake-bound and VC dimension

Learning: Core Idea

A learner receives experience and produces a model or hypothesis.

The model should perform well on unseen examples.

Learning involves representation, search/optimization and evaluation.

Generalization is the central objective.

Types of Learning

Supervised learning – labelled examples

Unsupervised learning – discover structure in data

Semi-supervised learning – labelled + unlabelled data

Reinforcement learning – learn through actions and rewards

Design of a Learning System

Define the task T

Define the experience E

Choose a performance measure P

Collect and prepare representative data

Select a representation and hypothesis space

Choose learning algorithm and evaluation procedure

T–E–P Formulation

Task (T): what the system must do

Experience (E): information available for learning

Performance (P): how success is measured

Example: spam classification → T = classify emails; E = labelled emails; P = accuracy/F1

Concept Learning

A concept is a Boolean-valued function over instances.

Positive examples satisfy the target concept.

Negative examples do not satisfy the target concept.

Goal: infer a hypothesis consistent with observed training examples.

Example: EnjoySport Concept

Attributes may include Sky, AirTemp, Humidity, Wind and Water.

Target concept: whether a day is suitable for a sport.

Training examples provide positive and negative instances.

The learner searches for a hypothesis that explains the examples.

Hypothesis

A hypothesis is a candidate description of the target concept.

A hypothesis maps instances to predicted class labels.

The hypothesis space H contains all hypotheses permitted by the representation.

Learning searches H for a useful hypothesis.

Hypothesis Space

H defines what the learner is capable of representing.

A richer H can represent more concepts but may increase search complexity.

A restricted H can simplify learning but may exclude the target concept.

Representation therefore affects learning performance.

Generalization and Specialization

Generalization covers more instances than the original hypothesis.

Specialization covers fewer instances.

Generalization is useful when a hypothesis is too specific.

Specialization is useful when a hypothesis is too general.

Version Space

Version space = set of hypotheses consistent with all training examples.

Positive examples eliminate hypotheses that reject them.

Negative examples eliminate hypotheses that accept them.

As training data increases, the version space generally becomes smaller.

Find-S: Basic Idea

Start with the most specific hypothesis.

Process positive examples.

Generalize only enough to cover each positive example.

Negative examples are not explicitly used by Find-S.

The result is one maximally specific consistent hypothesis under the representation.

Candidate-Elimination View

Maintains the set of hypotheses consistent with the data.

Uses a Specific boundary S and General boundary G .

Positive examples move S toward generalization and G toward consistency.

Negative examples specialize S/G boundaries as required.

Inductive Bias

Inductive bias = assumptions used to generalize beyond observed data.

Without bias, many hypotheses may fit the training data equally well.

Bias guides the search and determines which generalizations are preferred.

Examples: linearity, smoothness, simplicity, locality.

Restriction Bias vs Preference Bias

Restriction bias limits the hypotheses that can be represented.

Preference bias ranks or prefers some hypotheses over others.

Decision trees and linear models impose structural assumptions.

The bias affects both learnability and generalization.

Modelling in Machine Learning

Choose input representation X and target Y .

Define a model family or hypothesis class.

Estimate model parameters from training data.

Validate performance on data not used for fitting.

Use the final model for prediction or decision making.

Training, Validation and Test Data

Training set: used to fit the model.

Validation set: used for model/hyperparameter selection.

Test set: final unbiased performance estimate.

Data leakage occurs when information from evaluation data influences training decisions.

Overfitting and Underfitting

Underfitting: model is too simple to capture important structure.

Overfitting: model captures noise or peculiarities of training data.

Good generalization balances fit and complexity.

Regularization, validation and suitable representation help control overfitting.

Learning Frameworks

Frameworks formalize how learning performance and sample requirements are studied.

PAC learning studies probably approximately correct learning.

Mistake-bound models count prediction errors.

VC dimension characterizes expressive capacity of hypothesis classes.

PAC Learning

Probably: success holds with high probability.

Approximately correct: error is below a chosen tolerance.

Learning is evaluated using sample complexity and confidence/error parameters.

PAC analysis connects data requirements with learnability.

Mistake-Bound Model

Designed for online or sequential prediction settings.

Learner predicts before seeing the true label.

A mistake is counted whenever prediction differs from the true label.

Goal: derive an upper bound on the number of mistakes.

VC Dimension

VC dimension measures the capacity of a hypothesis class.

A set is shattered if every possible binary labelling can be represented.

Higher capacity can represent more patterns.

Capacity must be considered together with sample size and generalization.

Generalization Error

Training error measures performance on observed examples.

Generalization error measures expected performance on unseen examples.

The goal is low error on the underlying data distribution, not merely memorization.

Validation and test procedures estimate generalization empirically.

Learning Workflow

1. Define task and performance measure
2. Collect and preprocess data
3. Choose representation and hypothesis space
4. Train learner
5. Validate and tune
6. Evaluate on test data
7. Deploy and monitor

Worked Example: Concept Learning

Given labelled instances, identify attributes relevant to the target concept.

Start with a specific hypothesis.

Use positive examples to generalize where necessary.

Check consistency with negative examples.

The final hypothesis should balance consistency and useful generalization.

Key Takeaways

Learning is generalization from experience.

Concept learning uses examples to infer hypotheses.

Hypothesis space and inductive bias determine what can be learned.

Version spaces capture hypotheses consistent with observations.

PAC, mistake-bound and VC frameworks provide theoretical views of learning.

Chapter 3: Examination Focus

Define concept learning, hypothesis and hypothesis space.

Explain inductive bias with an example.

Solve a Find-S or version-space problem.

Differentiate training, validation and test sets.

Explain PAC learning, mistake-bound model and VC dimension.

Discuss overfitting and generalization.